Differential Gene Expression Analysis of Immune Cells exposed to ALS

Sushant Agellu | Saint Louis University | April 22, 2026

Abstract

Amyotrophic lateral sclerosis (ALS) is an incurable neuromuscular disease, recently appreciated as a systemic condition, with prominent immune peripheral contribution. This project reanalysed publicly available RNA-seq data from the GSE60424 dataset, which has paired, cell-sorted samples in multiple donors, to characterise transcriptional changes across circulating immune populations in ALS. Differential expression analysis between the samples of ALS patients (n = 3) and healthy controls (n = 4) was performed by cell-type strata — neutrophils, monocytes, CD4 T cells, CD8 T cells B cells, NK cells and whole blood — based on DESeq2 which fits a negative binomial generalized linear model then uses a Wald test with Benjamini–Hochberg correction after data cleaning and sample mapping.

Neutrophils and monocytes had some of the most robust transcriptional signatures with 38 and 34 differentially expressed genes (DEGs) respectively, across strata of expression size ≥5,045 to ≤9,637 genes retained per stratum (after filtering), while compartments from adaptive T-cells were largely quiescent in these tests (13–14 DEGs). Of these, 17 genes reached significance in neutrophils and monocytes and are candidates as common ALS-associated innate-immune markers. Principal component analysis on the top 2,000 most variable genes confirmed that cell-type identity, rather than disease status, dominated transcriptional variation, supporting the per-cell-type stratified analysis strategy . Residual seximbalance confounding not accounted for by this exploratory model was represented in the strongest ranked individual genes across cell types being dominated by sex-chromosome loci (DDX3Y, UTY, XIST, RPS4Y1).

Keywords: ALS, Celltypes,DESeq2, Bulk RNA Seq,Immune Cells

1. Introduction

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease characterized by the selective loss of upper and lower motor neurons, leading to muscle weakness, paralysis, and death within three to five years of symptom onset (Brown & Al-Chalabi, 2017). Although ALS was considered a central nervous system disorder, in recent years accumulating evidence has reframed this view to that of a systemic disease characterized by meaningful contributions from peripheral immune dysregulation to the progression of the disease. Altered myeloid-cell activation in ALS patients and SOD1 mouse models, increased circulating pro-inflammatory cytokines and Treg expansion (with reduced cytotoxic T-lymphocytes) all indicate that blood-accessible immune signals may indicate — or even influence — motor-neuron loss (Beers & Appel, 2019; Butovsky et al., 2012).

Thus, transcriptomic profiling of sorted immune cell populations provides a direct window into this peripheral compartment. Bulk whole-blood RNA-seq provides expression signals from many cell types, thereby diluting the signal of any particular cell type or obscuring signatures that are restricted to rarer populations. In contrast, cell-sorted RNA-seq derives homogenous populations prior to sequencing and

associates differential expression with a specific cell type. One of more useful public datasets available is the GSE60424 dataset (Linsley et al., 2014) which provides paired transcriptomes of neutrophils, monocytes, CD4 T cells, CD8 T cells, B cells and NK cell from donors with multiple autoimmune diseases and healthy controls alongside whole blood. This design is particularly well-suited to cell-type-stratified re-analysis in disease contexts where each cell subset may behave differently.

RNA-seq count data are integer-valued, discrete, strongly right-skewed and show overdispersion (i.e. variance greater than mean). As a result, methods based on the assumption of Gaussian or Poissondistributed data work very poorly in this problem. We do this through shrinkage of gene-wise dispersion via a negative binomial generalised linear model, which is ideal for small-sample RNA-seq studies (Love, Huber, & Anders 2014) and accounts for these statistical properties as we see in DESeq2. Together with proper quality-control visualisations (principal component analysis, sample distance heatmaps, dispersion plots) and effect-size diagnostics (volcano and MA plots), the DESeq2 pipeline provides an end-to-end template for count-based differential expression.

To complete this, the project had three clear research aims which were: (i) enumeration of differentially expressed genes (DEGs) in seven separate circulating immune cell populations from GSE60424 between ALS cases and healthy controls; (ii) what cell types carry the strongest transcriptional signature of ALS and how many genes are common across cell types as a measure of signal robustness; and, (iii) assess the methodological fitness for purpose of DESeq2 coupled with documenting statistical and biological flaws in existing data. We hypothesised that innate-immune populations (neutrophils and monocytes) would show a stronger ALS transcriptional signature than adaptive-immune populations (T and B cells), consistent with established inflammatory components of ALS pathobiology.

2. Methods

2.1 Data Source and Study Design

Raw count data and sample metadata for GSE60424 was downloaded from the NCBI Gene Expression Omnibus (Linsley et al., 2014). It consists of 134 RNA-seq libraries based on purified cell populations that were sorted from the peripheral blood of ALS, multiple sclerosis (pre- and post-treatment), type 1 diabetes, and sepsis donors or healthy control subjects utilized Illumina HiSeq instruments. In this cohort the comparison was limited to ALS-versus-healthy, resulting in three ALS donors and four healthy controls per cell type for the current study. Seven cell strata were queried in the study: neutrophils, monocytes, CD4 T cells, CD8 T cells, B cells, NK cells and whole blood. Sample counts by stratum are shown in Figure 1.

2.2 Data Management and Cleaning

The raw metadata and expression matrix were read with the data. References table package in R (version 4.5.0) New metadata column names were lowercased and punctuation-normalized to create machinereadable identifiers; for example, diseasestatus_ch1, celltype_ch1. We generated a sample-mapping table between GEO accessions (GSM IDs) and library IDs (lib221…lib355), and applied

consistent rownames (Ensembl geneIDs, ENSG format) on the count matrix. The output folder 01_DataCleaning contains flat CSVs of Cleaned inputs — a counts matrix (50,045 genes × 134 samples) and a sample map.

2.3 Statistical Analysis

Differential expression was performed using DESeq2 version 1.48.1 (Love et al., 2014). The lib IDs associated with ALS and healthy-control that corresponded to each of the seven cell strata were subsetted into count matrices. Genes with less than 10 counts in =2 samples), testing stratum contained between 7,215 - 9,637 genes. The design formula ~ disease was used to create a DESeqDataSet, where disease was coded as a two level factor with Healthy Control as the reference. Using R package DESeq2, we called the wrapper function DESeq() in default conditions with size-factor estimation by median-of-ratios method, gene-wise dispersion estimation with empirical-Bayes shrinkage toward a fitted mean–dispersion trend, and a Wald test for the disease coefficient. P-values . P-values were corrected for multiple testing with the Benjamini–Hochberg procedure. Genes were labelled as differentially expressed when padj < 0.05 and |log2 fold change| > 1.

Principled reasons for choosing DESeq2 over candidate alternatives Both the t-test and Wilcoxon rank-sum test assume a Gaussian distribution or other continuous data type, both of which are inappropriate for RNAseq I nteger counts. By assuming mean and variance are equal, the Poisson model breaks down on real sequencing data since biological replicates can vary. An alternative would have been the very similar negative binomial framework provided in edgeR (Robinson, McCarthy, & Smyth, 2010); however, we preferred DESeq2 because its dispersion-shrinkage estimator has recommended itself as more conservative particularly at the relatively small sample sizes available here (n = 3 vs. n = 4 per stratum), and consequently provides a lower false-positive rate when per-gene variance is poorly estimated. Limma-voom (Law et al., 2014) transforms counts to log-CPM and fits linear models; it performs best when each group has at least ten samples, which this study did not. A dispersion plots were produced for each cell type to verify that the negative binomial model fit was appropriate.

2.4 Normalization and Visualization

Counts were transformed based on variance-stabilizing transformation (vst) from DESeq2 only for visualization purposes to approximate homoscedasticity through the dynamic range of expression. PCA was performed on the 2,000 most variable genes (i) across all pooled ALS and healthy samples (global PCA), and separately within the neutrophil and monocyte strata (focused PCA). Volcano Plots, MA Plots, and the Top 30 DEGs Heatmaps per Focal Cell Type were Created Using ggplot2 and pheatmap. We constructed a consensus heatmap highlighting cross-cell-type signal DEGs by taking the intersection of significant DEGs (padj 1) in neutrophils and monocytes. We used the org. to map Ensembl IDs to HGNC gene symbols. Hs. Loading required package: illuminaHumanv4.dbBioconductor annotation package version 3.21.0.

3. Results

3.1 Study Design and Sample Composition

Restriction to ALS and healthy-control donors yielded a dataset of 49 libraries across seven cell strata (3 ALS and 4 healthy controls per stratum; Figure 1). One important step is that the sample balance was made uniform across cell types, which greatly simplifies downstream interpretation: all observed between-stratum differences in the counts of differential expression genes (DEGs) reflect biology rather than unequal statistical power from sample-size differences.

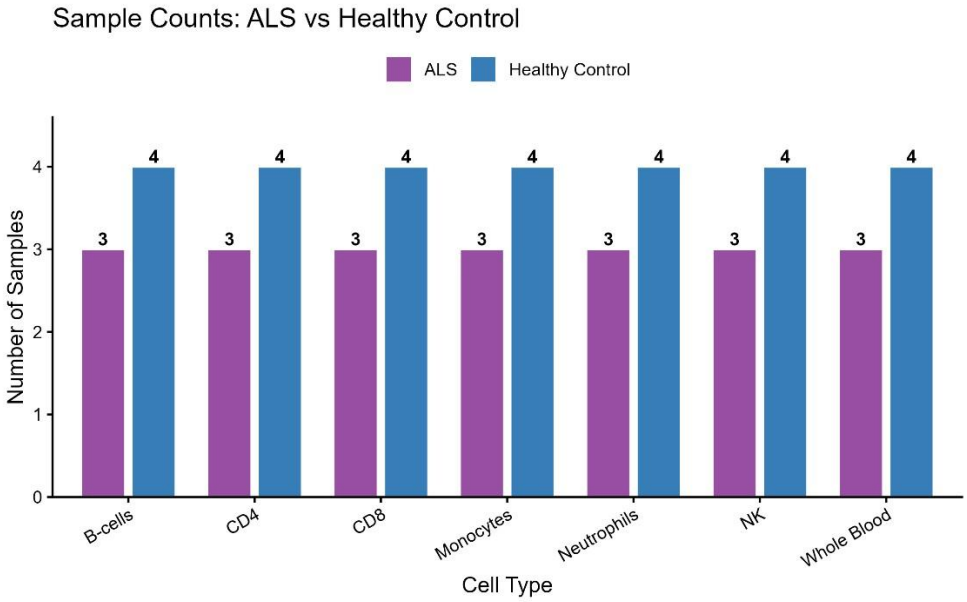

Figure 1. Sample counts per cell type for ALS (purple) and Healthy Control (blue) donors from GSE60424. Each cell type is represented by three ALS and four control donors, providing a balanced per-stratum design for differential expression testing. Numbers above each bar indicate sample counts.

3.2 Cell-Type Identity Dominates Transcriptional Variation

Principal component analysis (PCA) of the 2000 most variable genes across all ALS and healthy samples showed that the largest axes of variation correspond to celltype identity rather than disease state (Figure 2). Most of variance across the world was captured in PC1 and PC2 (Extended Data Fig. 1a) Samples from a specific cell type grouped tightly with one another irrespective of donor disease state. Thus, consistent with the expectation biologically (since transcriptomes of different immune cell populations are so divergent), this pattern confirms that a pooled analysis ignoring cell type would have been driven largely by differences in cell-type proportions rather than enrichment of disease signal. This result supported testing in a second DESeq2 model for each cell stratum.

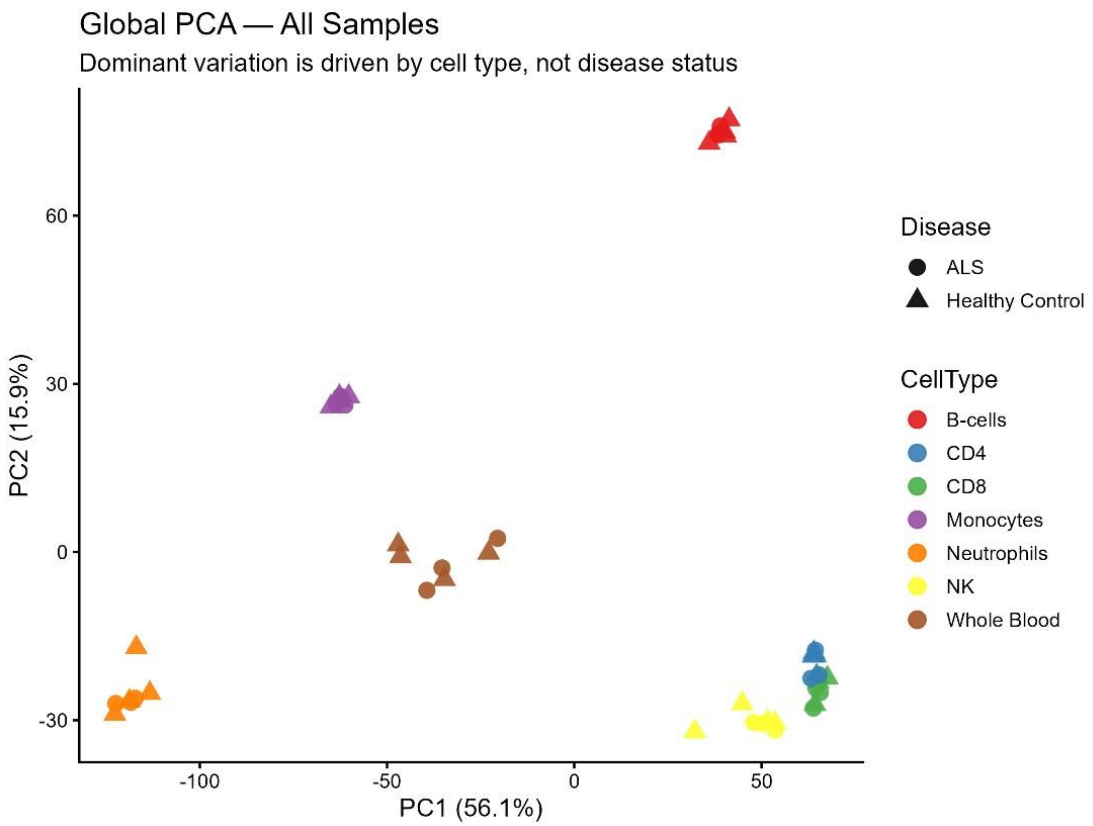

Figure 2. Global principal component analysis of all ALS and Healthy Control samples. Each point represents one RNA-seq library, colored by cell type and shaped by disease status (circle = ALS, triangle = Healthy Control). Tight clustering by color and mixing of shapes within each cluster indicates that cell-type identity, not disease status, is the dominant source of transcriptional variation. PC1 and PC2 are based on the variance-stabilized expression of the top 2,000 most variable genes.

3.3 Innate Immune Cells Show the Strongest ALS Signature

When analysing by DESeq2 it was clear and biologically relevant pattern: significantly more differentially expressed genes were produced from innate immune cells than adaptive immune cells (Figure 3, Table 1). The highest number of DEGs were found in neutrophils (38 in total; 10 up, 28 down), followed by monocytes (34; 18, 16) and NK cells (28). This yielded 25 DEGs in whole blood (containing multiple populations). By contrast, CD4 T cells (14), CD8 T cells (13) and B cells (14) demonstrated ~half as many changes, in line with the hypothesis that ALS exerts its peripheral transcriptional influence mainly through the innate arm of the immune-response system.

Table 1. DESeq2 differential expression summary per cell type, sorted by total DEG count. Filtering thresholds: padj < 0.05 and |log2 fold change| > 1. N_Samples reflects surviving libraries after matching to the count matrix; N_Genes is the number tested after low-count filtering (≥ 10 counts in ≥ 2 samples). Top_Gene is the most significant hit by adjusted p-value in each stratum.

Cell Type	No. of Samples	No. of Genes	Up	Down	Total DEGs	Top Gene

Neutrophils	7	7215	10	28	38	DDX3Y
Monocytes	7	8739	18	16	34	DDX3Y
NK	7	8968	16	12	28	XIST
Whole Blood	7	9627	9	16	25	RPS4Y1
CD4	7	8896	12	2	14	RPS4Y1
B-cells	7	8947	11	3	14	DDX3Y
CD8	7	8989	12	1	13	XIST

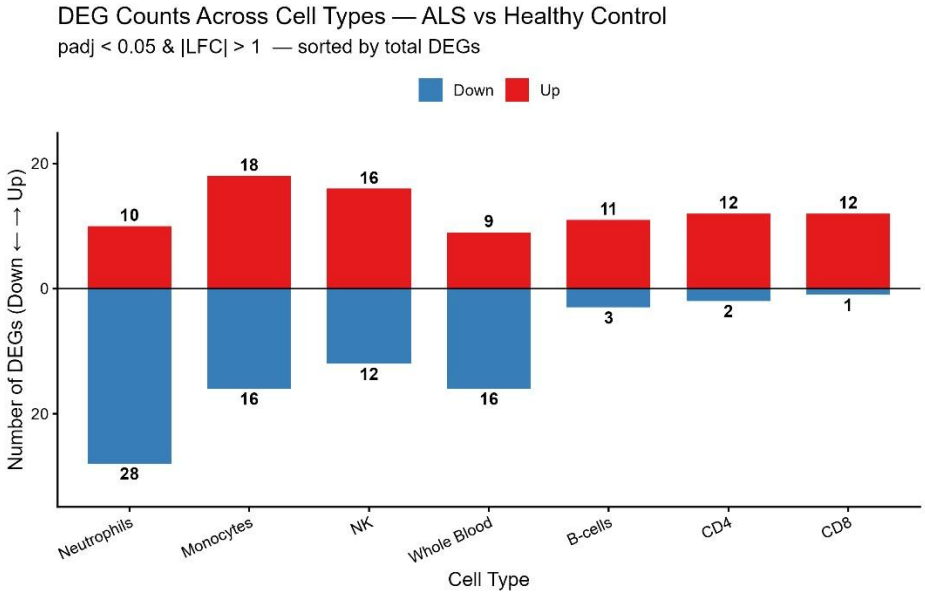

Figure 3. The number of differentially expressed genes per cell type in ALS compared to Healthy Control Counts of upregulated genes (red, positive direction) and downregulated genes (blue, negative direction), represented as a mirrored bar chart. Cell types are listed left-to-right based on total DEG count. Innate immune populations (ex. neutrophils, monocytes) predominate and adaptive lymphocytes (CD4, CD8, B cells) show significantly less signal. Abbreviations: (padj 1).

3.4 Focused Analysis of the Top Cell Types: Neutrophils and Monocytes

Principal component analysis restricted to neutrophils (Figure 4) and monocytes (Figure 5) showed partial but incomplete separation of ALS and healthy samples along the primary components, with 85% confidence ellipses overlapping — an expected outcome given small group sizes (n = 3 vs. n = 4) and modest effect sizes. Corresponding volcano plots (Figures 6 and 7) revealed a characteristic asymmetry in each cell type: a small number of strongly-changed genes with very small adjusted p-values, flanked by a larger population of sub-threshold genes. Sample-level heatmaps of the top 30 DEGs (Figures 8 and 9) showed that these top DEGs do create visually coherent blocks when samples are ordered by disease status, confirming that the

DEG set carries genuine sample-discriminating information rather than arising from a single outlier donor.

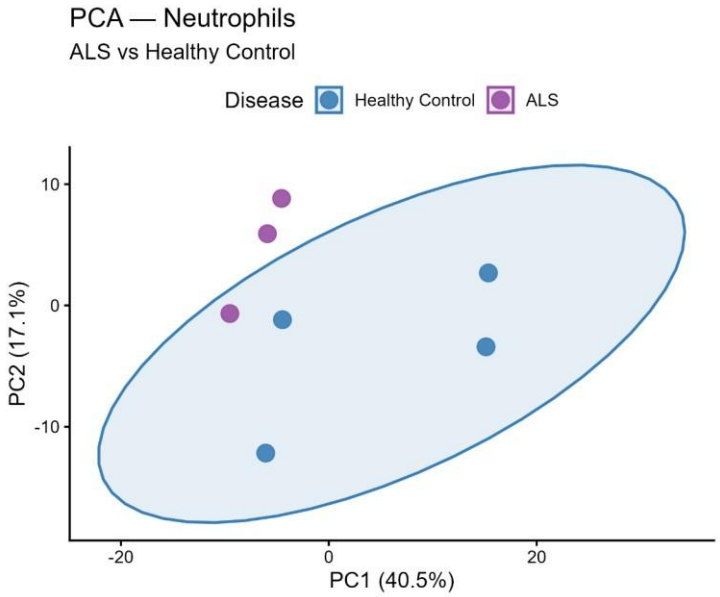

Figure 4. Focused principal component analysis of neutrophil samples, colored by disease status with 85% confidence ellipses. Variance-stabilized expression of the top 2,000 variable genes was used. Partial but incomplete separation on PC1 is observed between ALS (purple) and Healthy Control (blue) groups.

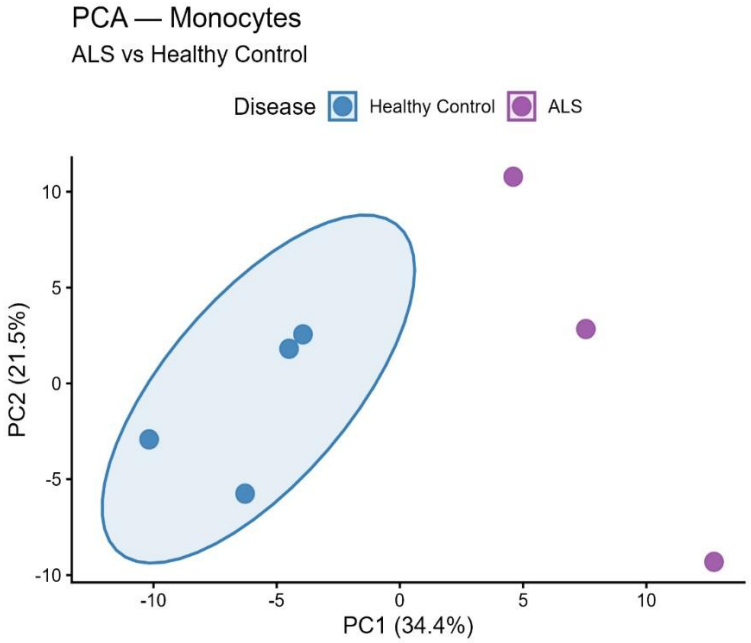

Figure 5. Focused principal component analysis of monocyte samples, constructed identically to Figure 4. The pattern mirrors the neutrophil result, with partial but overlapping clustering of disease groups.

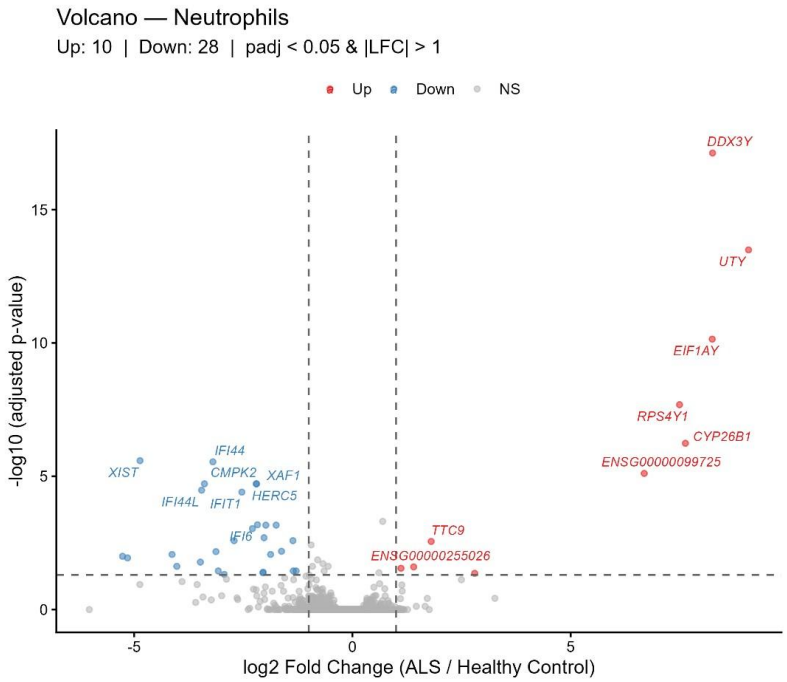

Figure 6. Differential expression in neutrophils (ALS vs. Healthy Control) Each point is a gene in this volcano plot. This shows log2 fold change on x-axis, -log10 adjusted p-value on y-axis. Red dots are over- (padj 1), blue dots are under-expressed (log2FC < -1) while grey dots represent genes showing no significant difference. The top eight upregulated and top eight down-regulated genes (by adjusted p-value are labeled with HGNC gene symbols.

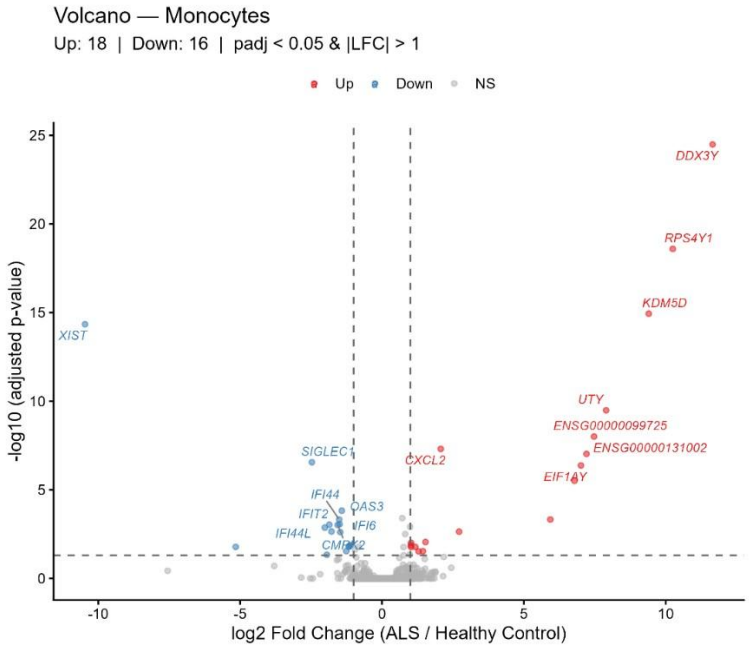

Figure 7. Volcano plot of differential expression in monocytes (ALS vs. Healthy Control), constructed identically to Figure 6.

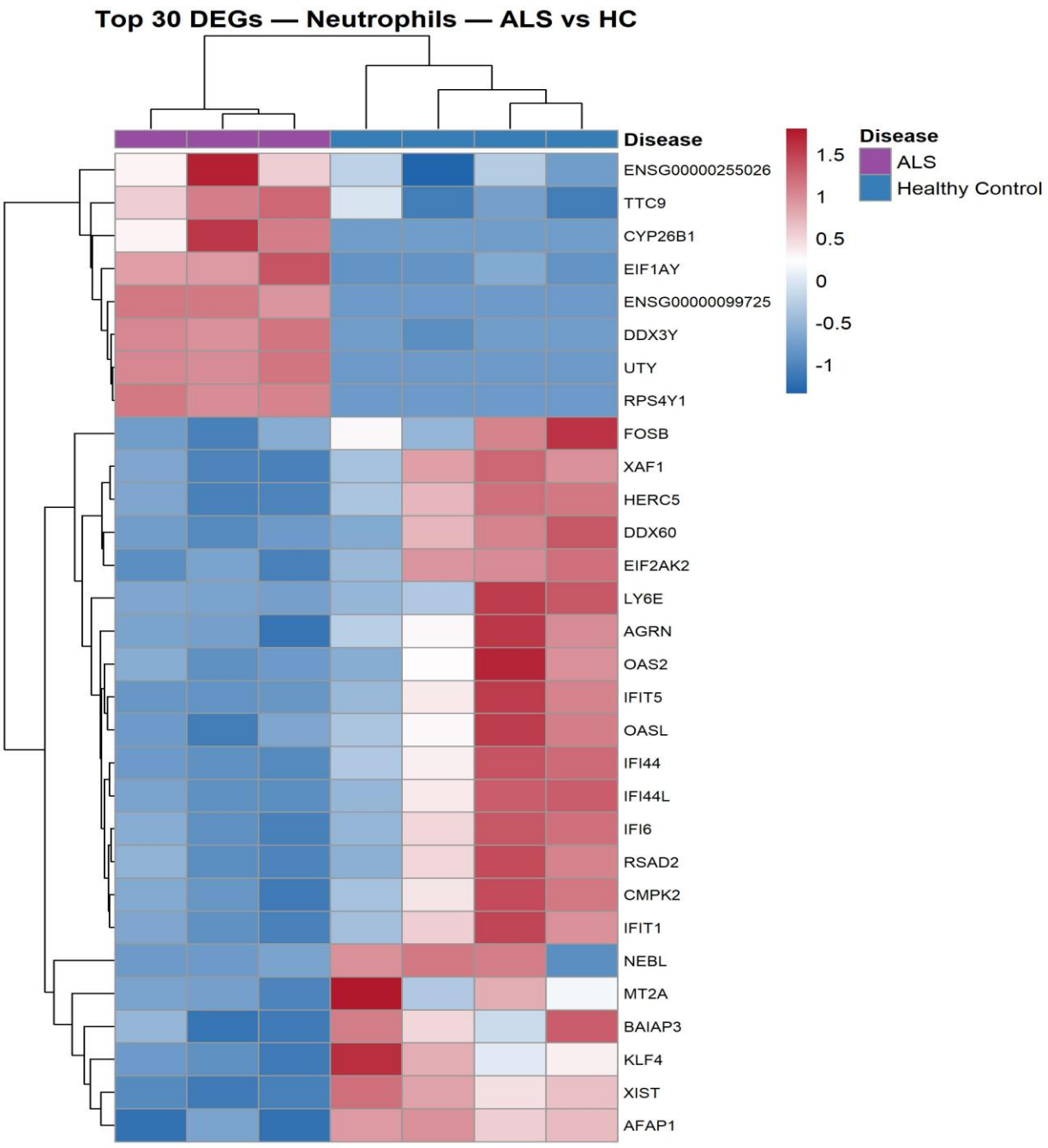

Figure 8. Top 30 DEGs in neutrophils (heatmap) Each row corresponds to a gene (with HGNC symbols) and each column represents an individual sample, grouped using colour shading by disease status (column annotation bar). Cell values are centered log2-transformed and z-scored percentage of variance-explained expression, with a ceiling at ±3. Red means larger than row mean expression; blue means smaller

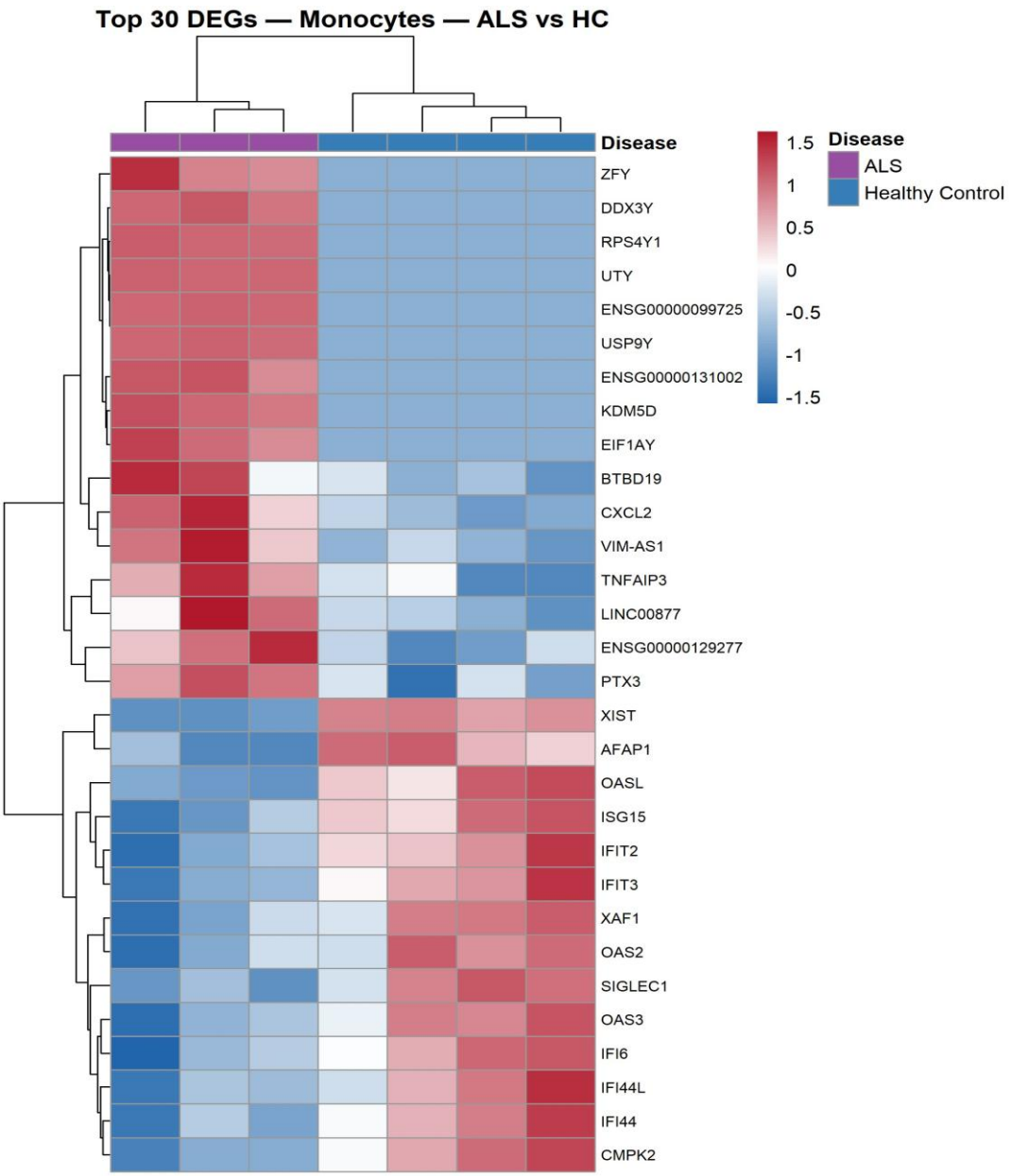

Figure 9. Heatmap of the top 30 differentially expressed genes in monocytes, constructed identically to Figure 8.

3.5 Consensus DEGs Shared Between Neutrophils and Monocytes

Intersection of DEGs significant in neutrophils and monocytes was used to identify robust candidate markers of ALS, which are also likely to generalize across innate-immune populations. A total of seventeen genes were retained in both cell types by using the combined passing criteria. Bottom row is the consensus heatmap (Figure 10), which shows log2 fold change of each shared gene in both to compare effect magnitude and direction. Some notions easily appeared: shared genes generally move in the same direction in both cell types, and effect sizes were widely similar. Selection for cross-cell-type concordance minimizes the chance that any of these signals is cell type specific artifact, and thus that makes seventeen individual genes the highest-confidence candidates emerging from this analysis.

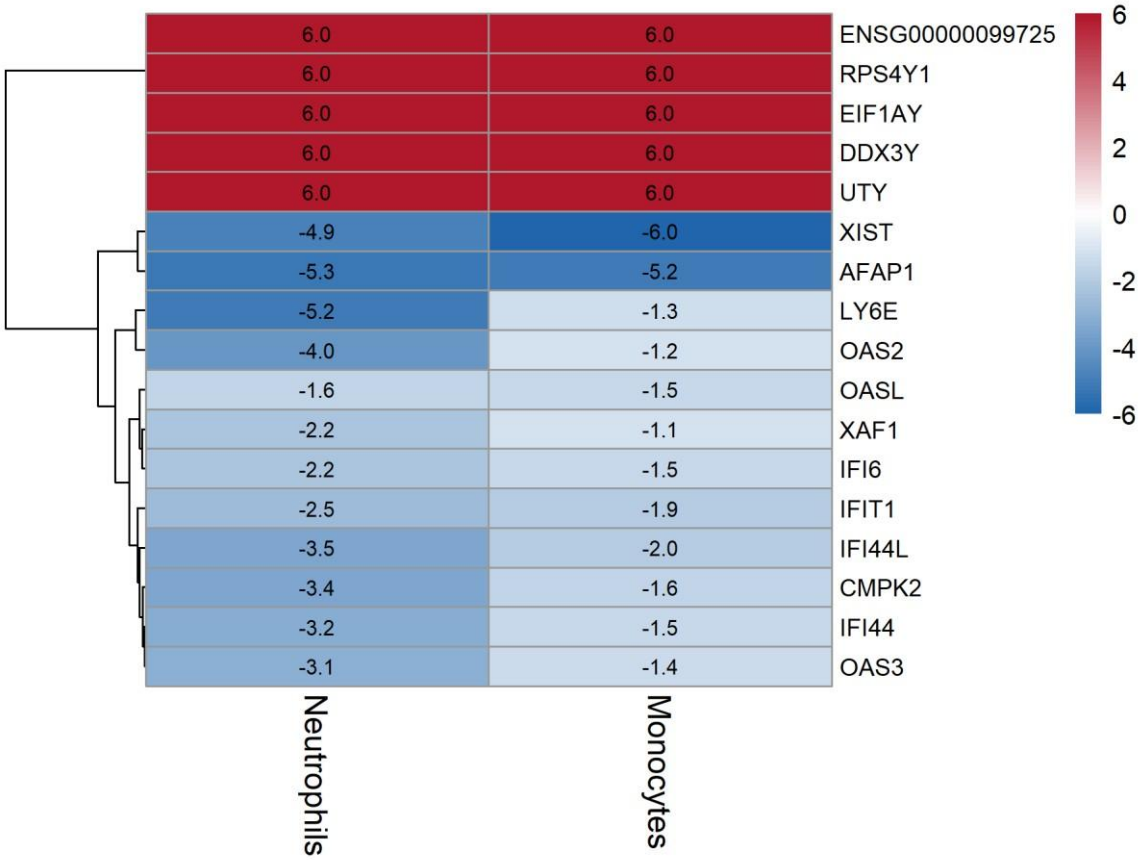

Figure 10. The consensus heatmap (from the sum of genes within each threshold padj 1 in neutrophils and monocytes) of significant differential expression data for the 17 genes significantly differentially expressed in neutrophils and monocytes is shown below. Cells exhibit log2 fold change (ALS / Healthy Control) per gene per cell type, where red corresponds to upregulation in ALS and blue signifies downregulation. The numbers inside each cell are the log2 fold change values. Rows are clustered, but columns are not — clustering would remove the ability to directly visually compare two or more columns.

The inspection of the top individual genes by cell type revealed that sex-chromosome-linked genes were ranked at the very top in each stratum; DDX3Y, UTY, KDM5D and RPS4Y1 on the Y chromosome and XIST on the X chromosome (Table 1, volcano plots). This pattern is diagnostic of a sex discrepancy in ALS and healthy-control groups: Y-linked genes, which are absent from the X-chromosome in females, differ stereotypically between male and female donors (as illustrated by top-ranked DEGs); more importantly still, with a cohort of only 3 versus 4 donors per stratum, even sparsely expressed single-donor sex imbalances could dominate the resultant list top-ranked DEG lists.

This does not mean the analysis is flawed, but that no gene with an absolute top ranks should be regarded as an ALS biomarker; however, the consensus DEGs (Figure 10) and overall counts of DEGs (Figure 3) are still biologically relevant.

4. Discussion

This re-analysis yielded three findings that both fit and expand on published work of peripheral immunity in ALS. First, the clear DEG-count asymmetry between cell types (innate-immune populations show about 2–3 times more DE genes than adaptive-immune populations) underpins a model of ALS-associated transcriptional change in the periphery that is dominated by the innate compartment. This is in agreement with observations of activated monocyte and neutrophil phenotypes in blood from patients with ALS (Beers & Appel, 2019; Murdock et al., 2017) and consistent with experimental evidence that loss of circulating monocytes may change motor-neuron death concentration across SOD1 mouse models (Butovsky et al., 2012).Second, these seventeen DEGs are especially useful because cross-cell-type concordance removes signals generated by technical artifact or rare-donor outliers unique to a single cell type; as such, they form a concise candidate list for validation in larger independent cohorts. Third, the PCA result—that cell type rules transcriptional diversity— is both a methodological check and an argument against pooled analysis of whole-blood in pursuit of disease-signal discovery in small cohorts given that confounds of cell-mixture will swamp any signals hidden within cell-type specific differences.

The most significant limitation of this analysis is the lack of adjustment for sex-covariates, as illustrated by many of the highest-ranked genes. The best hit for all seven cell-type DEG tables is a locus associated with the Y-chromosome or X-inactivation (Table 1), which is interpretable only as an artifact of sex-imbalance given small group sizes. An appropriate follow-up analysis would (a) limit to a gender-matched subset of donors, or (b) expand the design equation to ~ sex + disease such that sex-driven variability is regressed out prior testing for disease-negative coefficients. Due to the presence of only three ALS donors per cell type, neither option could be pursued in a cleaner fashion on this dataset thus framing the analysis as more exploratory.

First, with low statistical power for moderate-effect changes (n = 3 vs n = 4 per stratum), the counts of differentially expressed genes (DEGs) in Table 1 are a strong underestimate of the true number of expression changes associated with ALS within each cell type. Lastly, the top DEGs lists were not included in a pathway-level enrichment analysis, and integration of the DEG lists with gene-set databases (KEGG, Reactome, GO) would be an elegant next step to translate from gene level observations to biological hypotheses.

In conclusion, this project demonstrated an exploratory map of differential expression stratified by immune cell type based on publicly available peripheral immune datasets from ALS patients and healthy controls. We identified a core biological finding — an over-representation of innate-immune signature in neutrophils and monocytes, including seventeen consensus DEGs shared between them — that is consistent with the broader ALS immunology literature. In brief, the methodological key result is that whilst it is possible for DESeq2 to produce coherent, interpretable results with small cell-type-specific samples, this needs to be balanced against careful recognition of confounders (especially sex) and of statistical power.

The project is delivered as a reproducible numbered-folder workflow with every one analysis step implemented in a single commented R markdown script, corresponding to the level of detail/assurance needed for the specific course.

5. References

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